Native methods still Not Found in Table III (latest containers)

Totals: Cassandra 15, Jruby 7, Orientdb 8, Storm 26. All are methods declared native in a JAR shipped in the container. Search: Java_* exports of all 3,195 unique libraries under ~/Prototype, ~/Tutorial and ~/SysJava (40,622 exports), all 141 previous mapping results, method-name strings and RegisterNatives tables, the newest jffi and zstd-jni releases from Maven Central, and the web. None of these natives can execute in its container, so none can contribute syscalls.

Container	JAR	Class	Methods	#	Why it cannot be mapped	What was checked / source
cassandra_5.0.8	affinity-3.23.3.jar	net.openhft.ticker.impl.JNIClock	rdtsc0	1	Needs the native library of Chronicle's commercial Enterprise edition, which is not shipped and not publicly available; the code falls back to Java.	All Cassandra JARs (affinity-3.23.3, chronicle-*): no native library. Web: class ships in the open-source affinity artifact, native part only in the Enterprise edition (jar-download.com, net.openhft:affinity).
cassandra_5.0.8	affinity-3.23.3.jar	software.chronicle.enterprise.internals.impl.NativeAffinity	getAffinity0, getCpu0, getProcessId0, getThreadId0, rdtsc0, setAffinity0	6	Needs the native library of Chronicle's commercial Enterprise edition, which is not shipped and not publicly available; the code falls back to Java.	All Cassandra JARs (affinity-3.23.3, chronicle-*): no native library. Web: class ships in the open-source affinity artifact, native part only in the Enterprise edition (jar-download.com, net.openhft:affinity).
cassandra_5.0.8	jffi-1.3.11.jar	com.kenai.jffi.Foreign	compileNativeMethods, freeCompiledMethods, freeNativeMethod, invokeArrayWithObjectsReturnObject, newNativeMethod, registerNativeMethods, unregisterNativeMethods	7	Declared in jffi's Java class, but no jffi native library implements these methods (dead declarations); calling one throws UnsatisfiedLinkError.	Java_* exports of all 3,195 libraries on disk; strings and RegisterNatives in every jffi build on disk (1.3.9, 1.3.11, JRuby's); newest jffi 1.3.14-native from Maven Central: absent.
cassandra_5.0.8	zstd-jni-1.5.7-2.jar	com.github.luben.zstd.Zstd	generateSequences	1	Declared in zstd-jni's Java class, but not implemented in its native library (only zstd's internal ZSTD_ldm_generateSequences exists).	Shipped libzstd-jni-1.5.7-2.so and newest zstd-jni 1.5.7-6 (Maven Central): no Java_..._generateSequences.
jruby_10.1.0.0	jruby.jar	com.kenai.jffi.Foreign	compileNativeMethods, freeCompiledMethods, freeNativeMethod, invokeArrayWithObjectsReturnObject, newNativeMethod, registerNativeMethods, unregisterNativeMethods	7	Declared in jffi's Java class, but no jffi native library implements these methods (dead declarations); calling one throws UnsatisfiedLinkError.	Java_* exports of all 3,195 libraries on disk; strings and RegisterNatives in every jffi build on disk (1.3.9, 1.3.11, JRuby's); newest jffi 1.3.14-native from Maven Central: absent.
orientdb_3.2.53	jffi-1.3.9.jar	com.kenai.jffi.Foreign	compileNativeMethods, freeCompiledMethods, freeNativeMethod, invokeArrayWithObjectsReturnObject, newNativeMethod, registerNativeMethods, unregisterNativeMethods	7	Declared in jffi's Java class, but no jffi native library implements these methods (dead declarations); calling one throws UnsatisfiedLinkError.	Java_* exports of all 3,195 libraries on disk; strings and RegisterNatives in every jffi build on disk (1.3.9, 1.3.11, JRuby's); newest jffi 1.3.14-native from Maven Central: absent.
orientdb_3.2.53	jfr.jar	jdk.jfr.internal.JVM	setCompressedIntegers	1	Declared in the JDK 8 jfr.jar, but this JDK 8 libjvm registers no native for it (JFR backport differs from JDK 11+).	nm/strings of OrientDB's JDK 8 libjvm: no setCompressedIntegers / jfr_set_compressed_integers (only JfrOptionSet::compressed_integers).
storm_2.8.8	jline-2.14.6.jar	org.fusesource.hawtjni.runtime.Callback	bind, getEnabled, getEntryCount, getPlatform, reset, setEnabled, unbind	7	hawtjni-runtime classes bundled in jline-2.14.6.jar; hawtjni generates callback code into each project's own native library, and jline's libjansi was built without callbacks.	libjansi from jline-2.14.6 and all libraries on disk: no Java_org_fusesource_hawtjni_runtime_Callback_*. Web: hawtjni developer guide (callbacks are generated per project).
storm_2.8.8	rocksdbjni-10.10.1.1.jar	org.rocksdb.DBOptions	failIfOptionsFileError, setFailIfOptionsFileError	2	fail_if_options_file_error was deprecated in RocksDB 9.0 and removed from the C++ library; rocksdbjni 10.10.1.1 still declares the Java natives, its native library no longer implements them.	storm's librocksdbjni: absent; present only in an older rocksdb build (storm 2.4.0). Web: RocksDB 9.0 release notes, Kafka PR #21863.

Container	JAR	Class	Methods	#	Why it cannot be mapped	What was checked / source
storm_2.8.8	rocksdbjni-10.10.1.1.jar	org.rocksdb.LiveFileMetaData	newLiveFileMetaDataHandle	1	Declared in rocksdbjni 10.10.1.1, not exported by its native library.	storm's librocksdbjni and all libraries on disk: absent.
storm_2.8.8	rocksdbjni-10.10.1.1.jar	org.rocksdb.Options	failIfOptionsFileError, setFailIfOptionsFileError	2	fail_if_options_file_error was deprecated in RocksDB 9.0 and removed from the C++ library; rocksdbjni 10.10.1.1 still declares the Java natives, its native library no longer implements them.	storm's librocksdbjni: absent; present only in an older rocksdb build (storm 2.4.0). Web: RocksDB 9.0 release notes, Kafka PR #21863.
storm_2.8.8	rocksdbjni-10.10.1.1.jar	org.rocksdb.RocksEnv	disposeInternalJni	1	Declared in rocksdbjni 10.10.1.1, not exported by its native library.	storm's librocksdbjni and all libraries on disk: absent.
storm_2.8.8	rocksdbjni-10.10.1.1.jar	org.rocksdb.WBWIRocksIterator	refresh0Jni	1	Implemented, but exported only under a C++-mangled name (missing extern "C" upstream), so the JVM cannot link it.	storm's librocksdbjni exports _Z46Java_org_rocksdb_WBWIRocksIterator_refresh0JniP7JNIEnv_P8_jobjectl, not Java_org_rocksdb_WBWIRocksIterator_refresh0Jni.
storm_2.8.8	storm-shaded-deps-2.8.8.jar	org.apache.storm.shade.io.netty.handler.codec.quic.Quiche	quiche_enable_debug_logging	1	Shaded netty quiche class; the native library registers netty's unshaded class names, so the shaded native cannot link.	Same check as the shaded io_uring/snappy libraries.
storm_2.8.8	storm-shaded-deps-2.8.8.jar	org.apache.storm.shade.org.xerial.snappy.BitShuffleNative	shuffle, shuffleDirectBuffer, unshuffle, unshuffleDirectBuffer	4	Shaded (relocated) class. The bundled libsnappyjava.so exports only unshaded Java_org_xerial_snappy_* names, so the JVM cannot link the shaded natives (UnsatisfiedLinkError). Relocating snappy-java breaks its JNI binding.	storm-shaded-deps-2.8.8.jar's libsnappyjava: 19 Java_org_xerial_snappy_* exports, 0 shaded. Web: SPARK-36681, 'snappy-java as a native library should not be relocated'.
storm_2.8.8	storm-shaded-deps-2.8.8.jar	org.apache.storm.shade.org.xerial.snappy.SnappyNative	arrayCopy, isValidCompressedBuffer, maxCompressedLength, nativeLibraryVersion, rawCompress, rawUncompress, uncompressedLength	7	Shaded (relocated) class. The bundled libsnappyjava.so exports only unshaded Java_org_xerial_snappy_* names, so the JVM cannot link the shaded natives (UnsatisfiedLinkError). Relocating snappy-java breaks its JNI binding.	storm-shaded-deps-2.8.8.jar's libsnappyjava: 19 Java_org_xerial_snappy_* exports, 0 shaded. Web: SPARK-36681, 'snappy-java as a native library should not be relocated'.